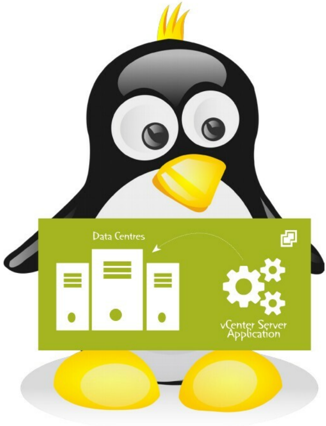
An Introduction to govcsim (a vCenter Server Simulator)

govcsim is a vCenter Server and ESXi API based simulator that offers a quick fix solution for prototyping and testing code. It simulates the vCenter Server model and can be used to create data centres, hosts, clusters, etc.



govcsim (a vCenter Server simulator) is an open source vCenter Server and ESXi API based simulator written in the Go language, using the *govmomi* library. govcsim simulates the vCenter Server model by creating various vCenter related objects like data centres, hosts, clusters, resource pools, networks and datastores.

If you are a software developer or quality engineer who works with vCenter and related technologies, then you can use govcsim for fast prototyping and for testing your code.

In this article, we will write an Ansible Playbook to gather all VMs installed on a given govcsim installation. Ansible provides many modules for managing and maintaining VMware resources. (You can find out more about Ansible modules for managing VMware at *http://docs.ansible.com/ansible/list_of_cloud_modules.html#vmware.)* Do note that govcsim will simulate almost the identical environments provided by VMWare vCenter and ESXi server.

Installation

We will use Fedora 26 for the installation of govcsim. Let's assume that Ansible has been already installed using *dnf* or a source tree.

The requirements for installing govcsim are:
1. Golang 1.7+
2. Git

Step 1: Installing Golang

To install the Go tools, type the following command at the terminal:

```
$ sudo dnf install -y golang
```

Step 2: Configuring the Golang workspace

Use the following commands to configure the Golang workspace:

```
$ mkdir -p $HOME/go
$ echo 'export GOPATH=$HOME/go' >> $HOME/.bashrc
$ source $HOME/.bashrc
```

Check if everything is working by using the command given below:

```
$ go env GOPATH
```